

Repository Mirroring Between GitHub and GitLab Using Git

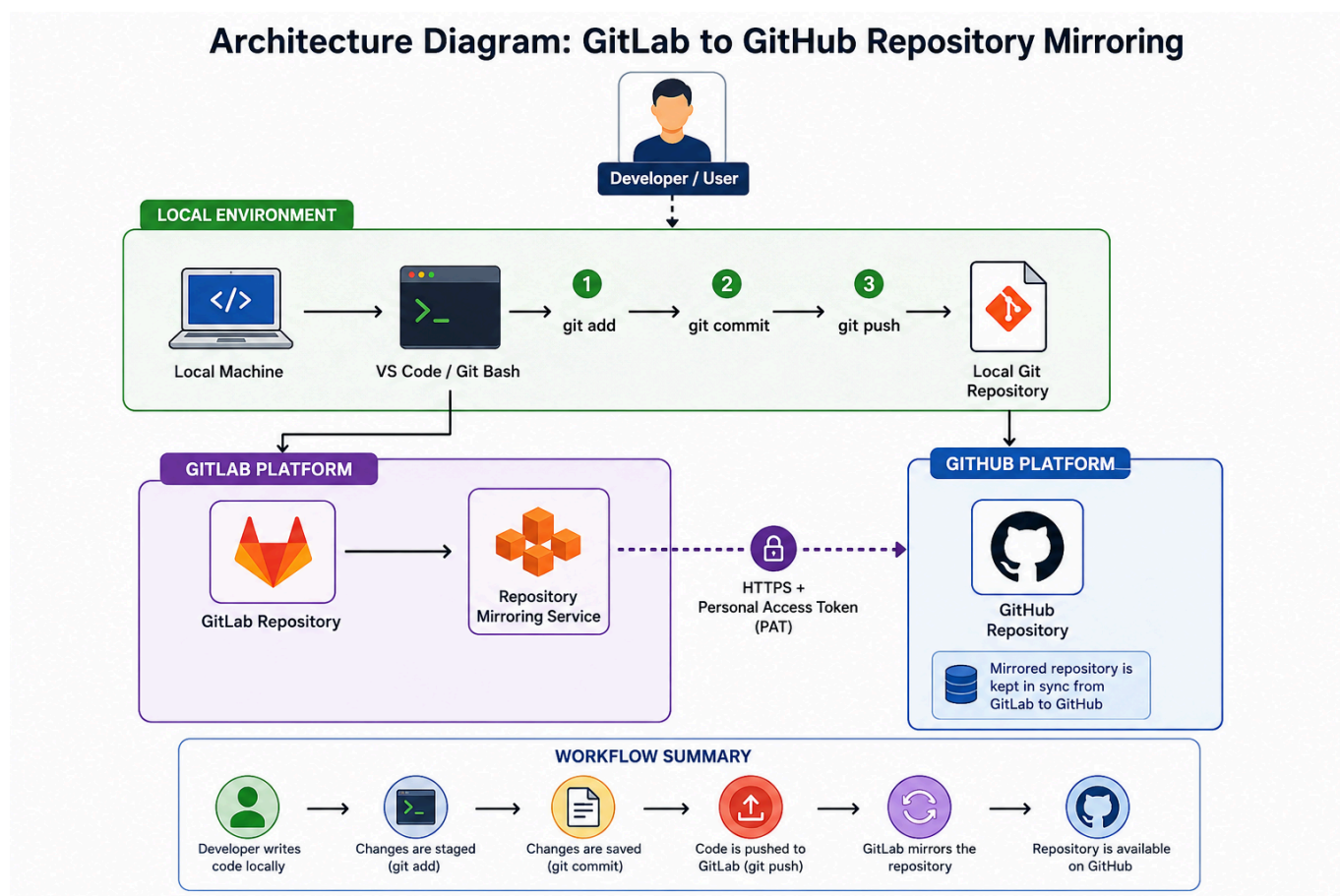
Introduction

This project demonstrates a repository mirroring workflow where code developed on a local machine is synchronized with both GitHub and GitLab using multiple Git remotes. Changes made in the local repository are pushed to both platforms, ensuring that each repository remains consistent and up to date. This implementation provides a practical approach to maintaining identical codebases across multiple Git hosting platforms through manual repository mirroring.

This approach is commonly used for:

- Backup repositories
- Multi-platform availability
- Professional Git workflows

Architecture Diagram



Tools and Platforms Used

- Local Machine (Windows)
- Git Bash
- VS Code

- GitHub
- GitLab

Step-by-Step Mirroring Workflow

Step 1 : Create Repository on GitHub

1. Login to GitHub
2. Click New Repository
3. Enter repository name
4. Create repository

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository.](#)
Required fields are marked with an asterisk (*).

1 General

Owner * pranalirirgithub / Repository name * mirror-github
✓ mirror-github is available.

Great repository names are short and memorable. How about [stunning-parakeet](#)?

Description
0 / 350 characters

2 Configuration

Choose visibility * Public
Choose who can see and commit to this repository

Add README
READMEs can be used as longer descriptions. [About READMEs](#)
On ☒

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)
No .gitignore

[Create repository](#)

2 Configuration

Choose visibility * Public
Choose who can see and commit to this repository

Add README
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On ☒

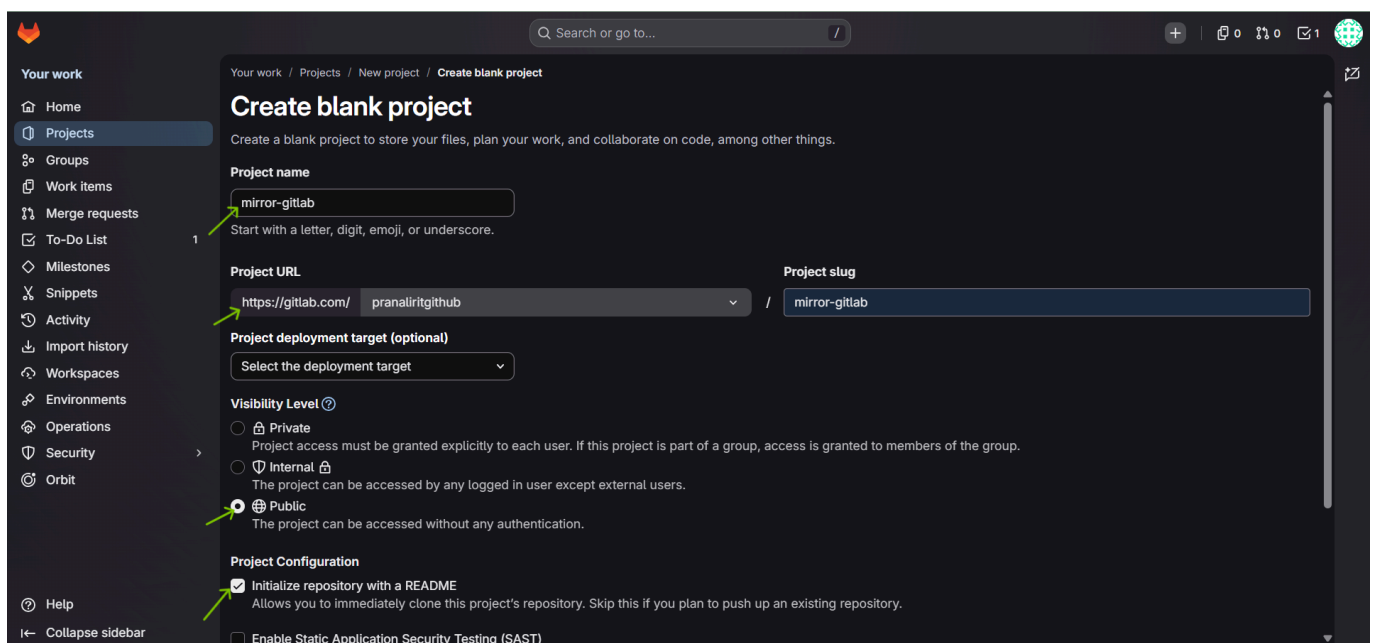
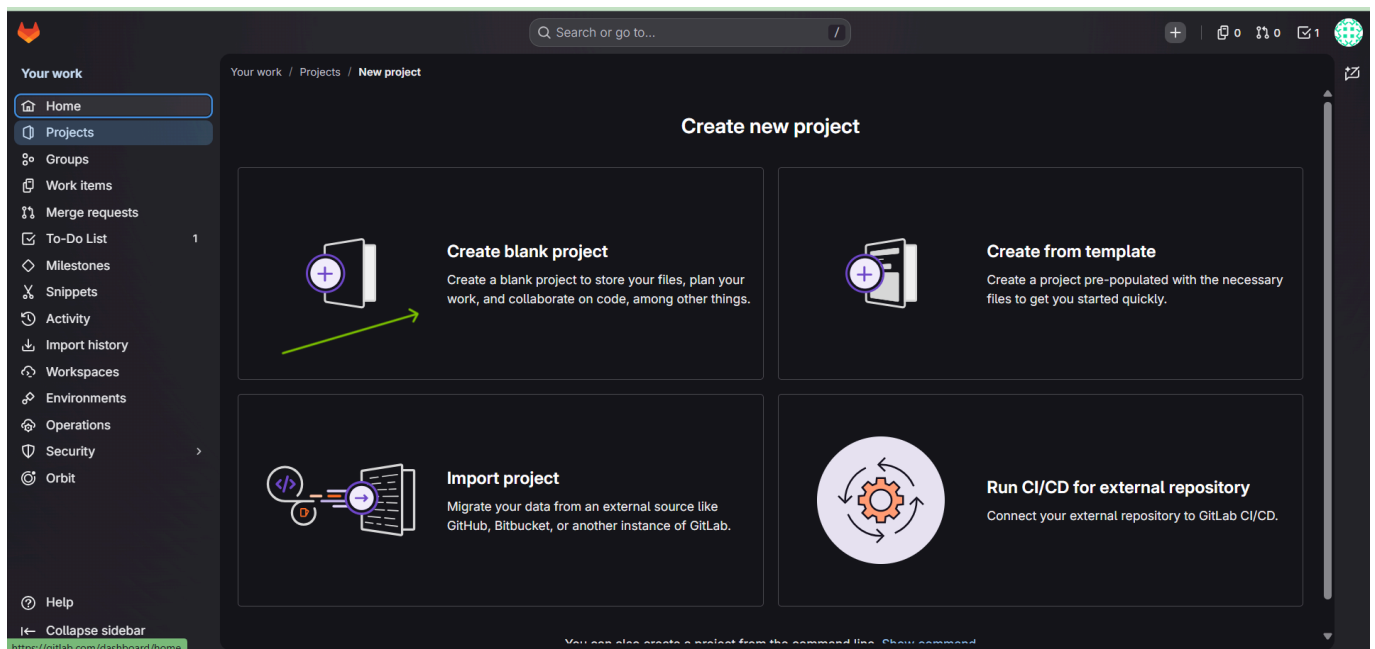
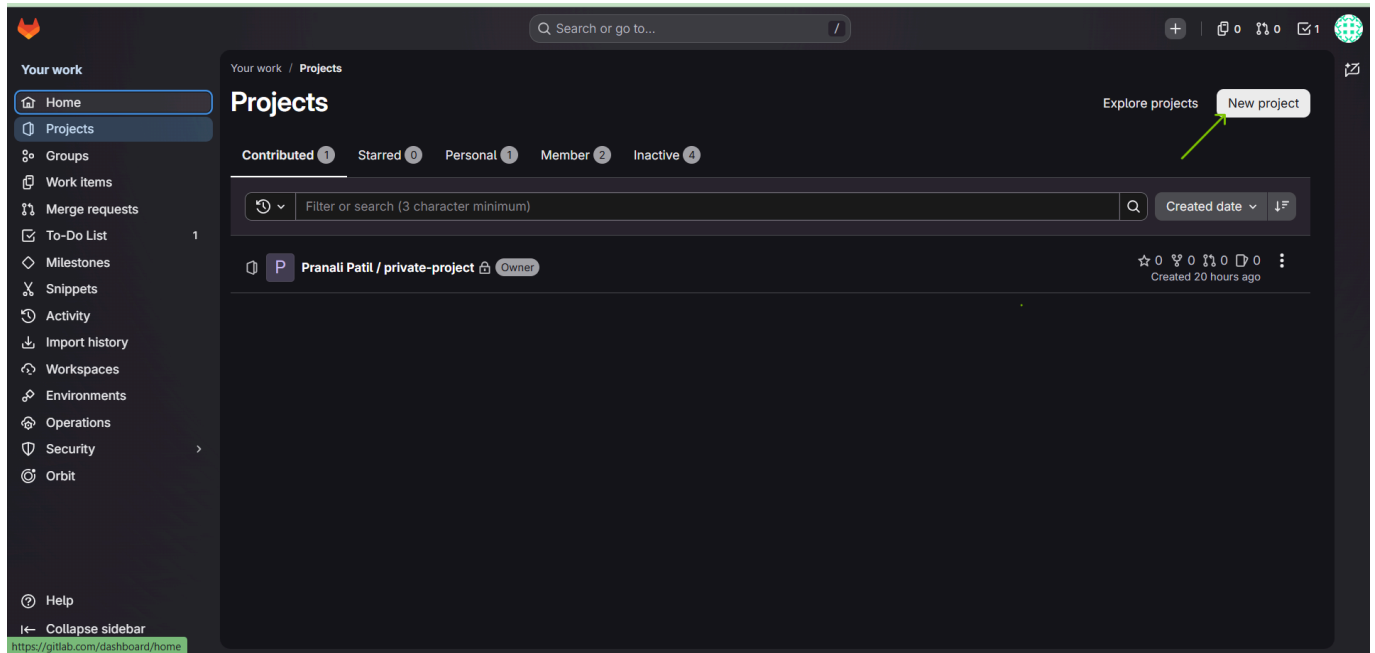
Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)
No .gitignore

Add license
Licenses explain how others can use your code. [About licenses](#)
No license

[Create repository](#)

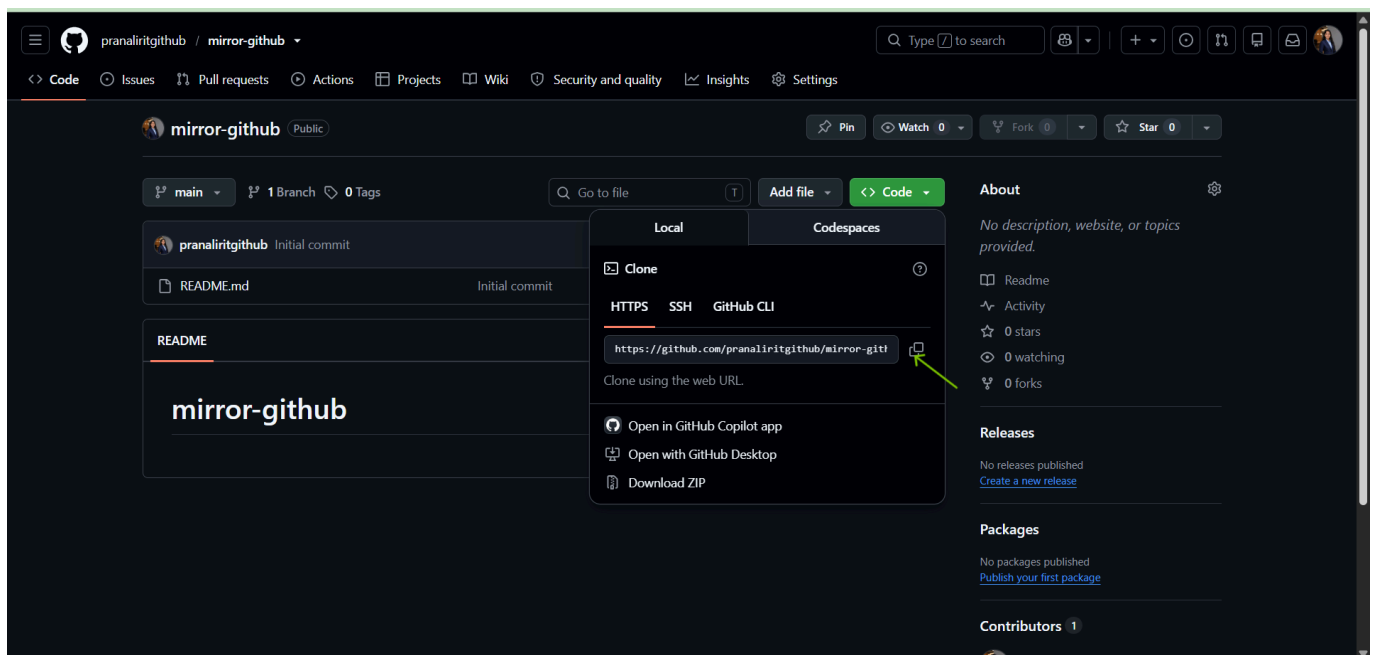
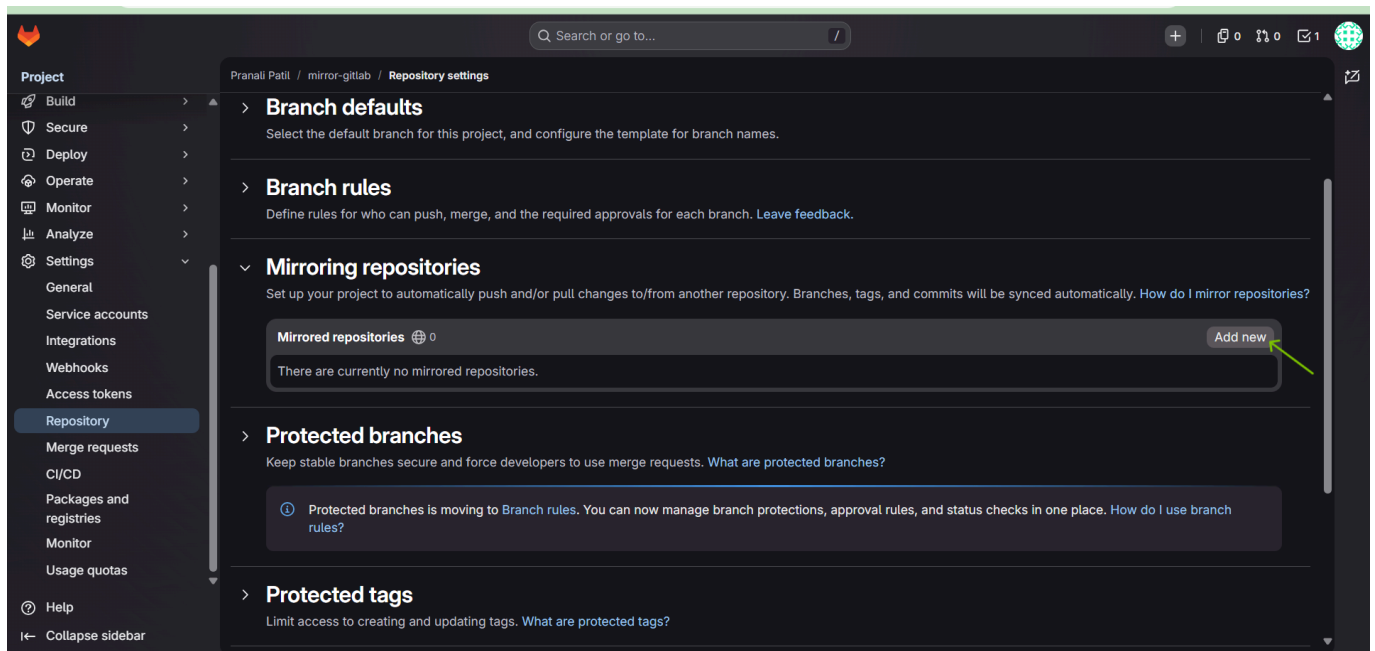
Step 2 : Create Repository on GitLab

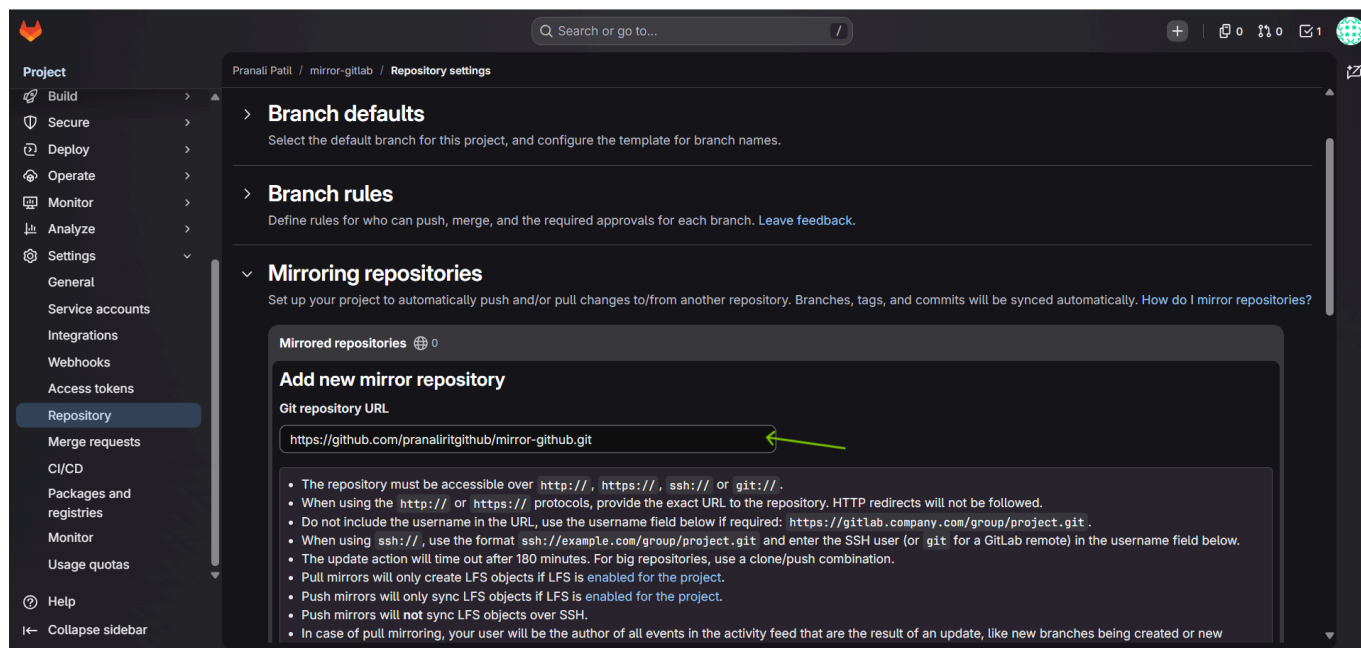
1. Login to GitLab
2. Click New Project / Repository
3. Enter repository name
4. Create repository



Step 3 : Configure Repository Mirroring in GitLab (HTTPS URL)

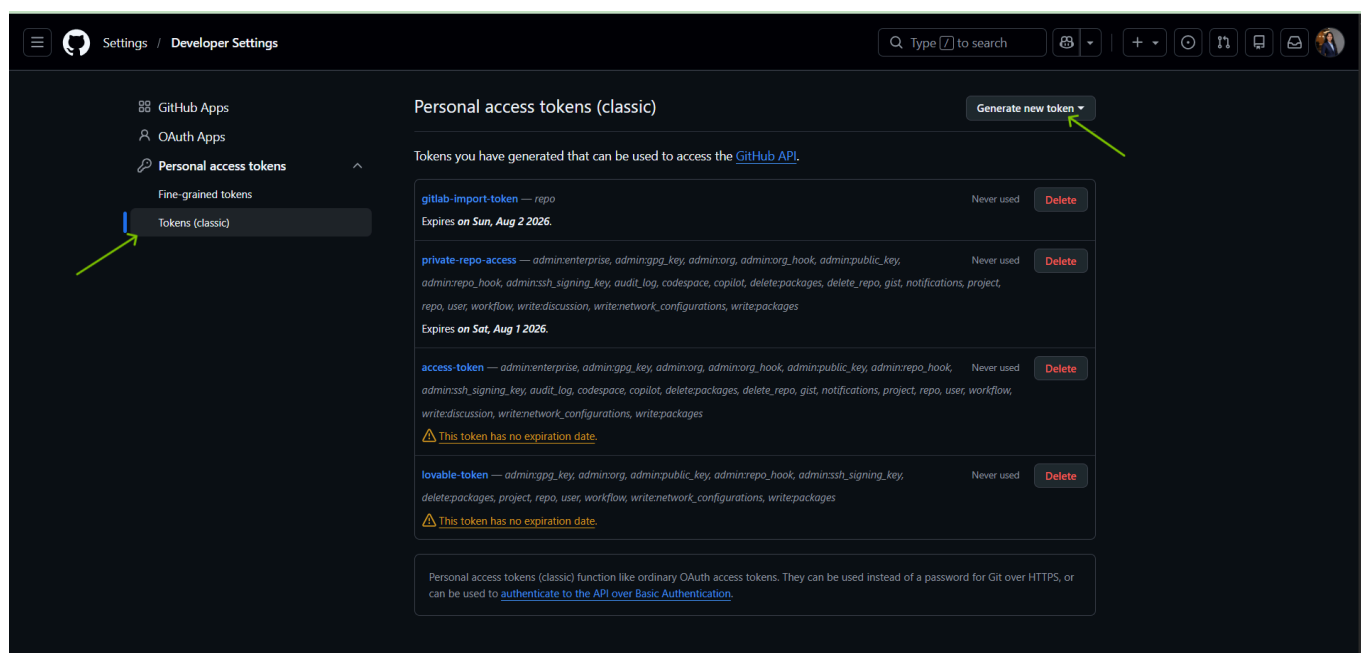
1. Open your GitLab repository
2. Go to Settings → Repository
3. Scroll to Mirroring repositories
4. Select add new
5. Paste the GitHub repository HTTPS URL
6. Save the mirror configuration

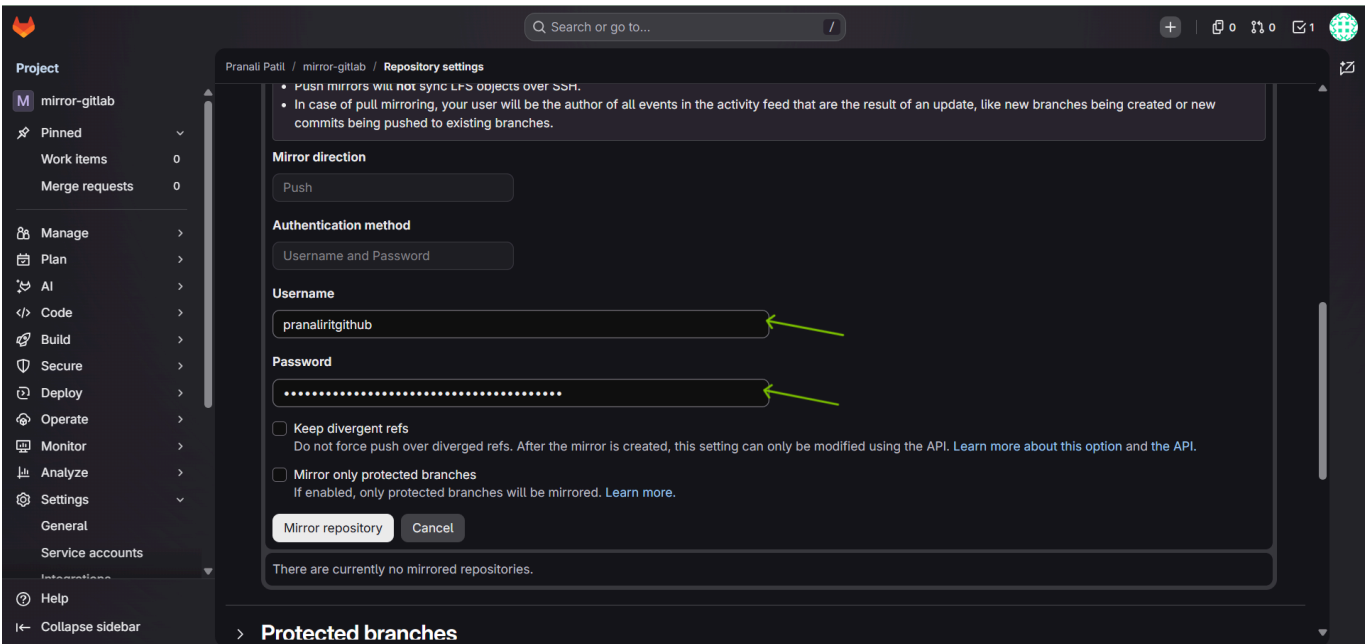
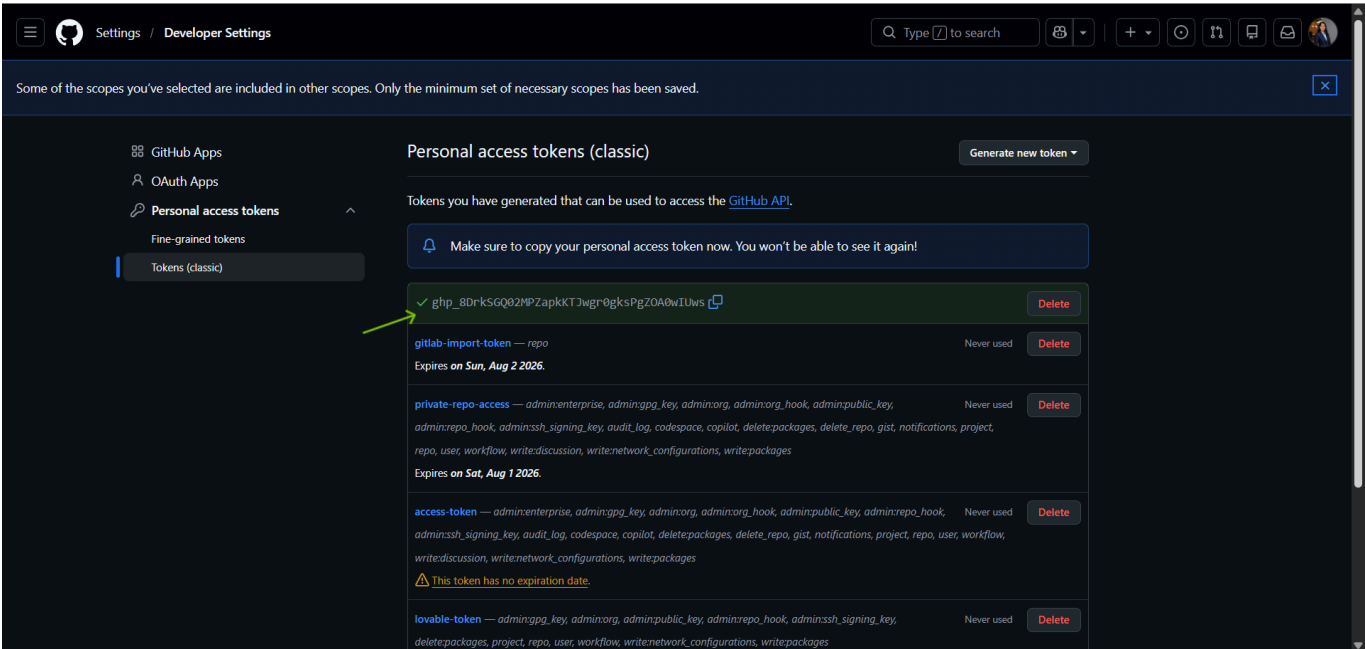
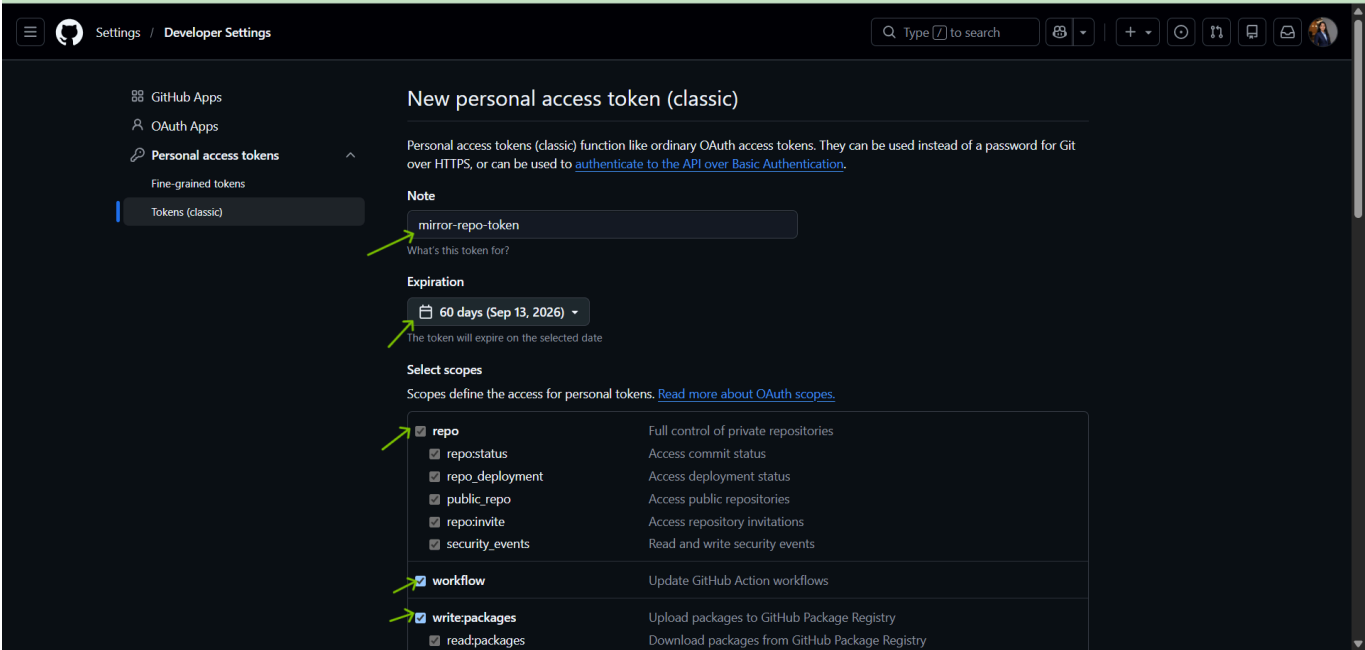




Step 4 : Create GitHub Personal Access Token

1. Login to GitHub
2. Go to Settings → Developer settings
3. Open Personal access tokens
4. Select Generate new token (Classic)
5. Give a token name
6. Set token expiration (limit days as required)
7. Set token visibility as private
8. Select all required permissions (check all boxes)
9. Generate the token, copy it, and paste this token into the GitLab repository mirroring authentication section

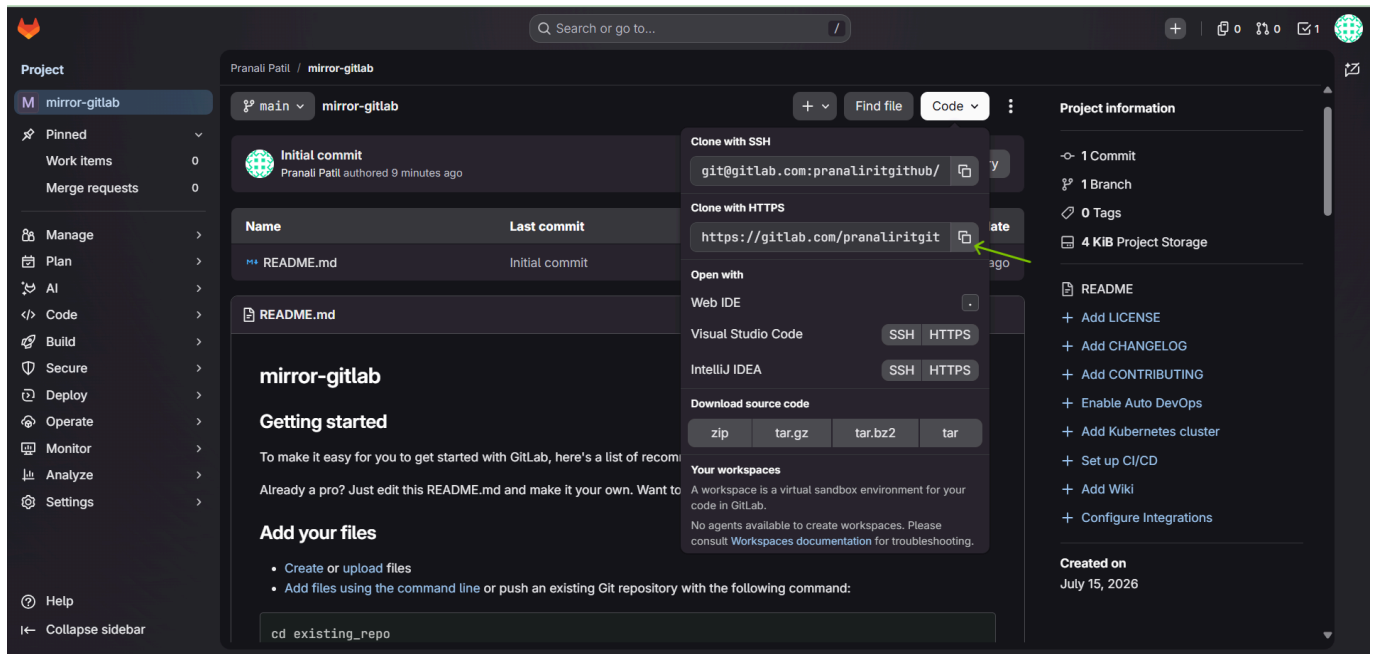




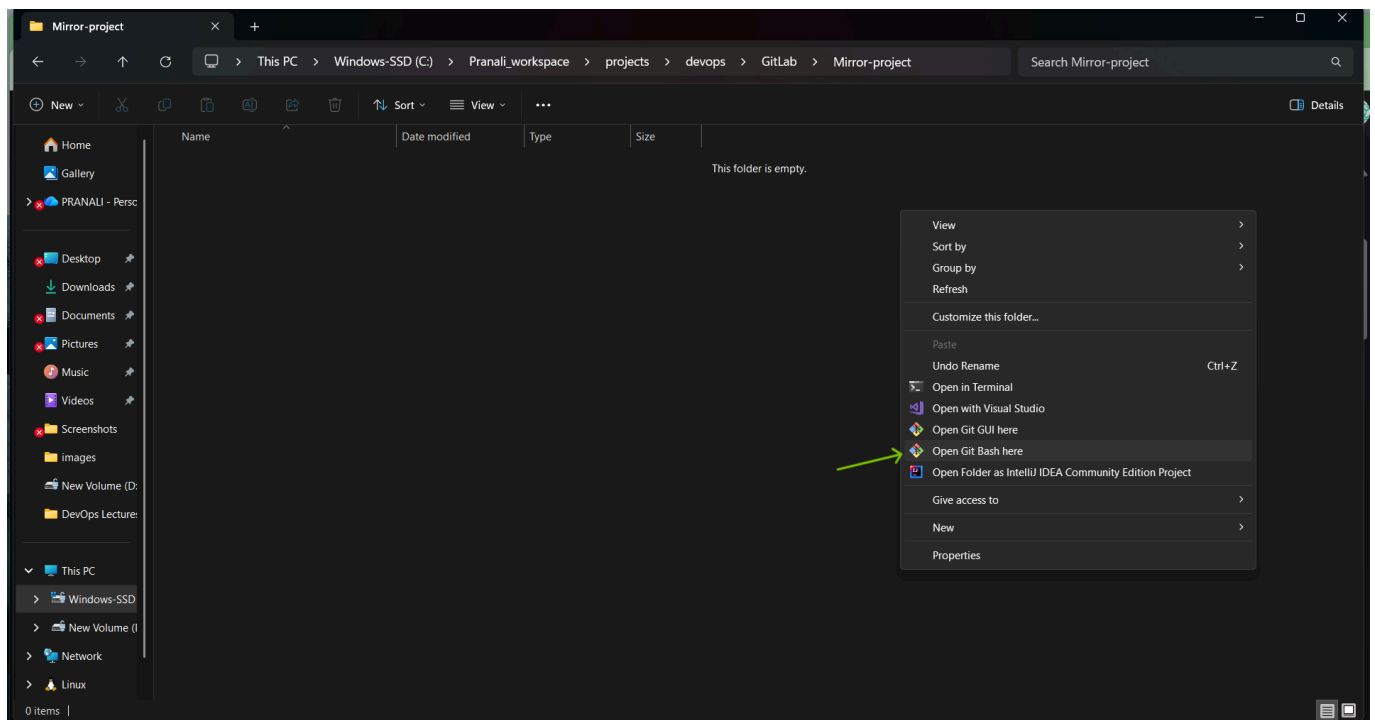
Step 5 : Clone One Repository to Local Machine

You can clone either GitHub or GitLab repository

```
git clone https://gitlab.com/username/repo-name.git
cd repo-name
```



- Right-click inside the folder, click Show more options, then select Open Git Bash here to open Git Bash in this directory




```
MINGW64~/c/Pranali_workspace/projects/devops/GitLab/Mirror-project/mirror-gitlab

prati@LAPTOP-QENHHIN3 MINGW64 /c/Pranali_workspace/projects/devops/GitLab/Mirror
-project
$ git clone https://gitlab.com/pranaliritgithub/mirror-gitlab.git
Cloning into 'mirror-gitlab'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

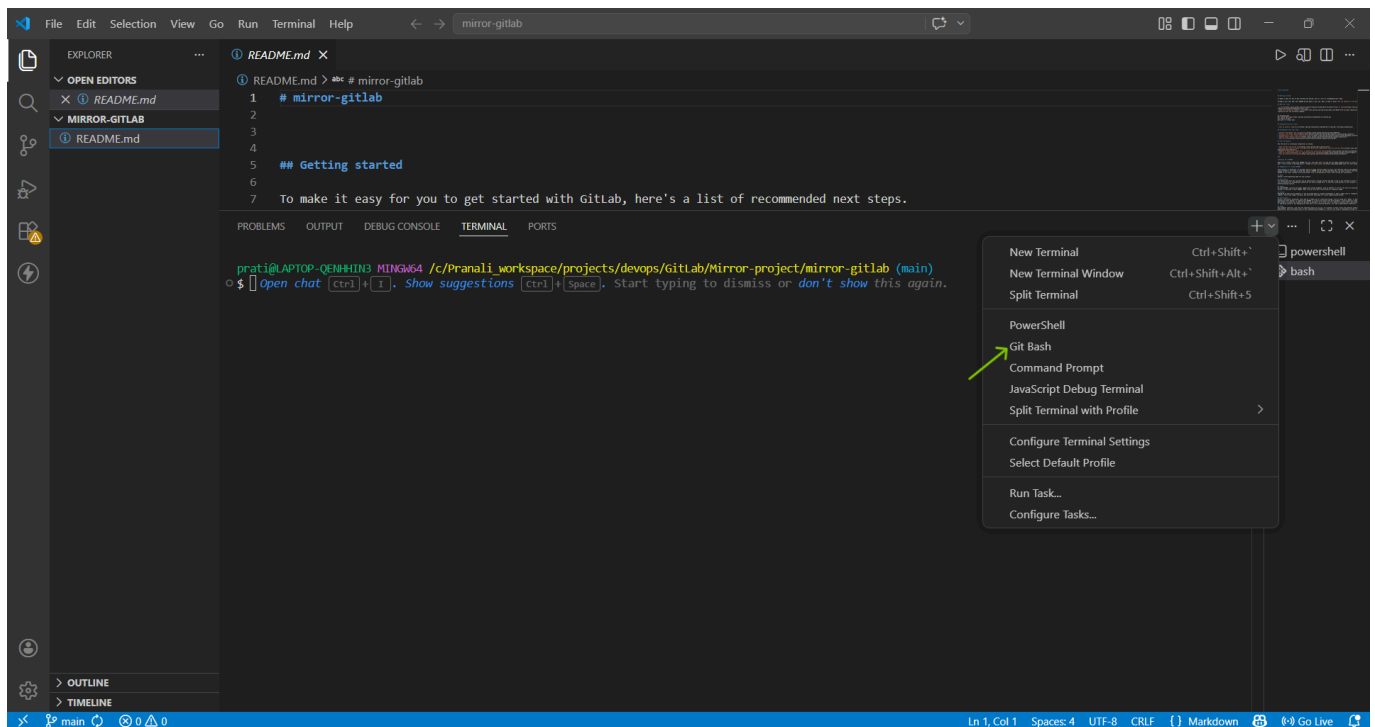
prati@LAPTOP-QENHHIN3 MINGW64 /c/Pranali_workspace/projects/devops/GitLab/Mirror
-project
$ ls
mirror-gitlab/

prati@LAPTOP-QENHHIN3 MINGW64 /c/Pranali_workspace/projects/devops/GitLab/Mirror
-project
$ cd mirror-gitlab/

prati@LAPTOP-QENHHIN3 MINGW64 /c/Pranali_workspace/projects/devops/GitLab/Mirror
-project/mirror-gitlab (main)
$ code . ; exit
```

Step 6 : Open Project in VS Code

```
code . ; exit
```



Step 7 : Create or Modify Files Locally

Examples:

- index.html
- README.md

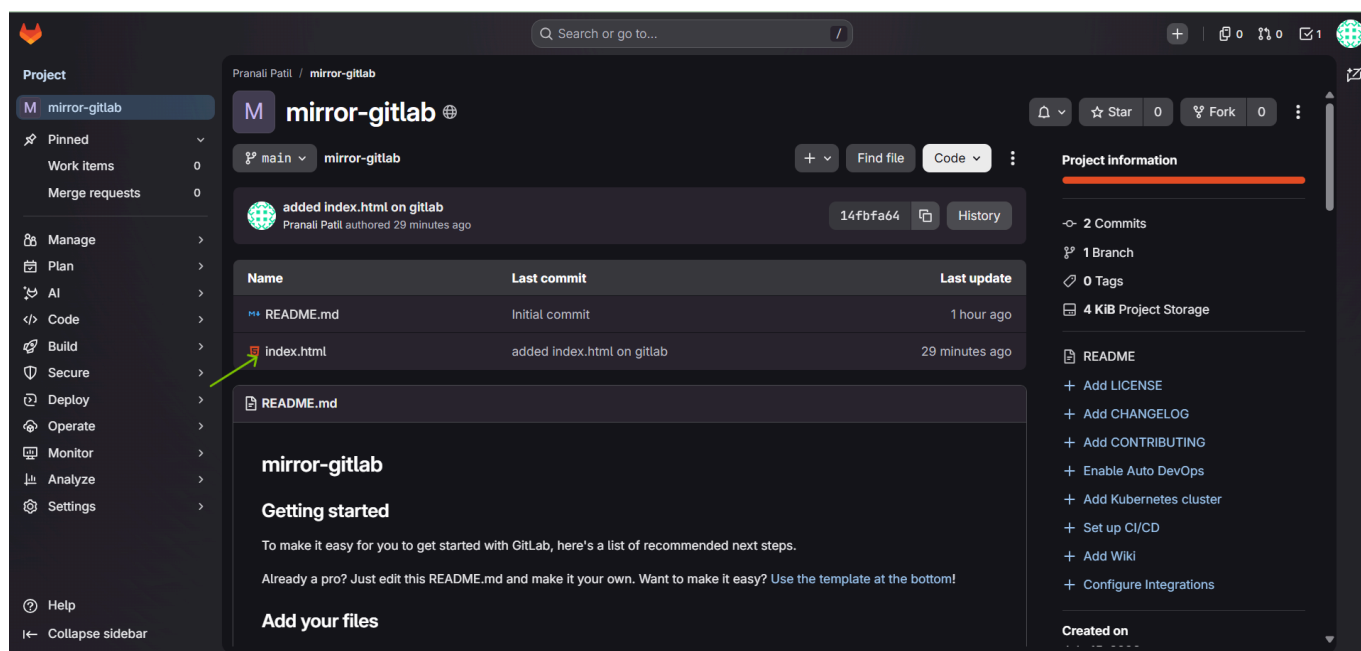
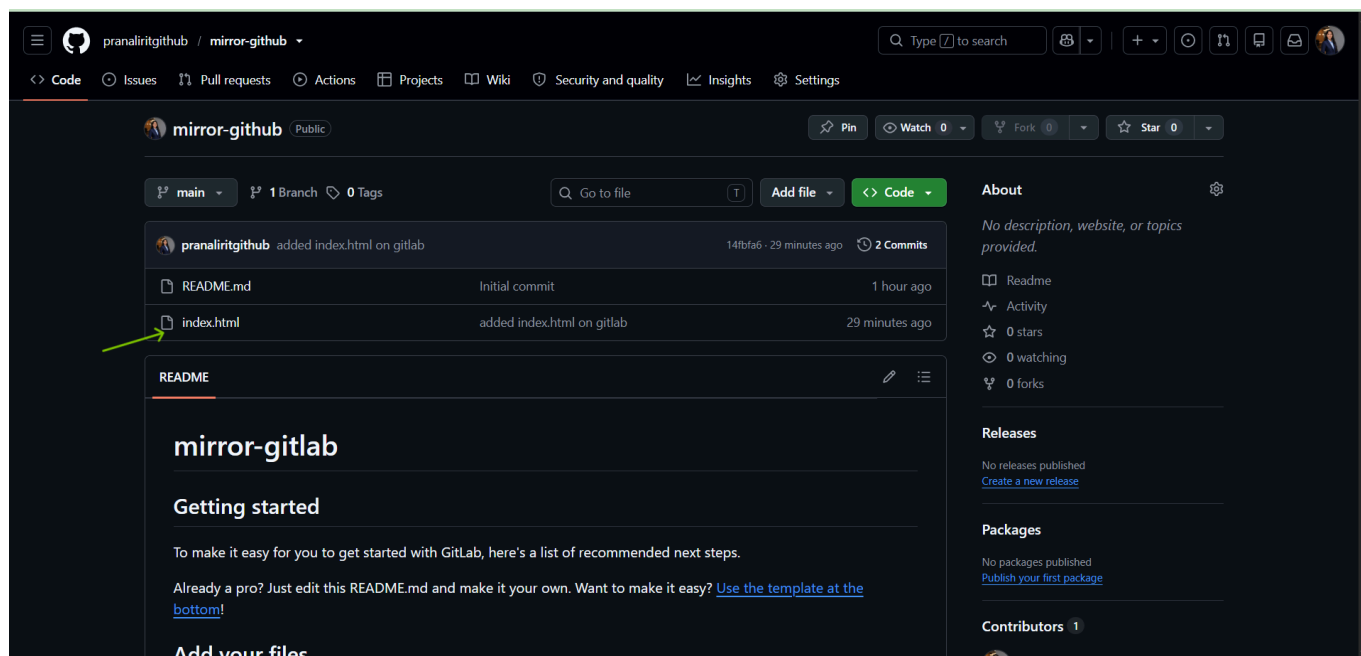
Stage and Commit Changes

```
git status git add . git commit -m "Commit for mirroring to GitHub and GitLab"
```


Step 8 : Push File into Gitlab and Github

```
git push -u origin main
```





Summary

This project demonstrates a repository mirroring workflow that synchronizes source code between GitHub and GitLab from a single local Git repository. The project is developed locally using Git and Visual Studio Code, while both remote repositories remain synchronized through GitLab Repository Mirroring and multiple Git remotes.

The implementation includes creating repositories on GitHub and GitLab, configuring GitLab Repository Mirroring using the GitHub HTTPS repository URL, generating a GitHub Personal Access Token (PAT) for secure authentication, and managing version control operations using Git. Code changes committed locally are pushed to both platforms, ensuring that the repositories remain consistent and up to date.

This solution enhances code redundancy, simplifies multi-platform repository management, and demonstrates practical Git workflows and repository synchronization techniques commonly adopted in modern software development and DevOps environments.